New Minor Subject – Digital Humanities

A student, taking up a new subject in their second year, takes 20 ECTS in a new subject. They then choose at the end of Year 2 one of the following options:

- Continue this subject to a Minor award, taking a further 20 ECTS in the subject in both Junior and Senior Sophister years.
- Continue as breadth in JS year, taking 10-ECTS in the minor in JS year, and discontinuing study of the subject at the end of that year.
- Discontinue the study of the minor altogether.

Overview Detail

Capacity for New Minor Subject	No fixed capacity		
Primary contact for queries	Head of Department	Secondary contact for queries	
Pre-requisite/Eligibility Requirements for Admission: None			

Modules available at Level 1

- Students on the New Minor Subject will take these in their Year 2
- Must total 20 ECTS, 10 ECTS in each semester

Module Code[1]	Module	ECTS Value	Semester of Delivery (1, 2 or Year-long)	Please indicate and pre- or co-requisites (Module Code)	Mandatory or Optional	Notes[2]
DHXXX	Introduction to Digital Humanities: Text, Data and Interpretation	5 ECTS	1		Mandatory	For students with no prior knowledge of DH
DHXXX	Digital Methods and Computation	5 ECTS	1		Mandatory	For students with no prior

	al Thinking for the Humanities					knowledge of DH
DHXXX	Digital Archives, Cultural Heritage and Historical Knowledge	5 ECTS	2	Pre-Digital Methods and Computational Thinking for the Humanities	Mandatory	
DHXXX	Data Collection and Corpus Building for Humanities Research	5 ECTS	2	Pre-Digital Methods and Computational Thinking for the Humanities	Mandatory	

--- Introduction to Digital Humanities: Texts, Data and Interpretation (5 ECTS)

[Content: What is DH; How do humanities materials become data; Digitization and knowledge production; Digitization of literary, historical, and philosophical materials.]

[Case studies: literary corpus; historical archives; philosophical texts]

[Key question: what changes occur in meaning when text is converted into data]

--- Digital Methods and Computational Thinking for the Humanities (5 ECTS)

[Content: computational thinking; algorithmic reasoning; Automated research processes]

[Technologies: Python basics; data cleaning; simple scripts]

[How is technology changing research methods?]

--- Digital Archives, Cultural Heritage and Historical Knowledge (5 ECTS)

[Content: digital archives; metadata systems; cultural heritage digitisation]

[Case studies: national archives; library databases; museum digitization]

[Tools: Omeka… ; Dublin Core metadata]

[Key question: when history is digitized, who decides what to preserve?]

--- Data Collection and Corpus Building for Humanities Research (5 ECTS)

[Content: literary corpus construction; web scraping; data cleaning]

[Technologied: Python_BeautifulSoup, requests]

[Questions: data bias; corpus selection problem]

Modules available at Level 2

- Students in the New Minor Subject will take these in their Year 3
- Must total 20 ECTS, 10 ECTS in each semester
- Please note in the comments section which modules should be made available as 10-ECTS breadth where a student wishes to continue the studies in the subject, but not proceed to an award.

Module Code[1]	Module	ECTS Value	Semester of Delivery (1, 2 or Year-long)	Please indicate and pre- or co-requisites (Module Code)	Mandatory or Optional	Notes[2]
DHXXX	Computational Text Analysis and Distant Reading	5 ECTS	1	Pre-Data Collection and Corpus Building for Humanities Research	Mandatory	
DHXXX	Spatial Humanities: Mapping History and Literature	5 ECTS	1		Mandatory	

DHXXX	AI, Algorithms and the Humanities	10 ECTS	2		Mandatory	

--- Computational Text Analysis and Distant Reading (5 ECTS)

[Content: distant reading; stylometry; topic modelling; sentiment analysis]

[Tools: Python; NLTK; spaCy]

[Case studies: literary studies. __ analyzing Ulysses's character network, narrative structure, and lexical distribution.]

[Questions: does distance reading weaken literary interpretation?]

---Spatial Humanities: Mapping History and Literature (5 ECTS)

[Content: spatial humanities; literary cartography; historical GIS]

[Technologies: QGIS; Gephi]

[Case studies: mapping migration history; literary geography__Dublin literary spaces: urban spaces in Joyce's works

[Questions: does spatial data simplify historical narratives?]

---AI, Algorithms and the Humanities (10 ECTS)

[AI and knowledge production: how machine learning generates knowledge. Automated explanation vs. human explanation]

[Algorithmic power: algorithmic bias; data inequality; platform power]

[Culture and literature issues: AI-generated text and literary creation; the evolution of the concept of authorship]

[Historical and archival issues: the application of AI in historical research; the risks of automated archival analysis]

[Philosophical discussion: human interpretation vs. algorithmic interpretation; can data represent reality?]

[technical section as case studies: simple machine learning; NLP model examples]

The focus remains on theory and critique.

Modules available at Level 3 or 4

- Students on the New Minor Subject will take these in their Year 4

Module Code[1]	Module	ECTS Value	Semester of Delivery (1, 2 or Year-long)	Please indicate any pre- or co-requisites (Module Code)	Mandatory or Optional	Notes[2]
DHXXX	Technology, Ethics and Digital Society	10 ECTS	1		Mandatory	
DHXXX	Digital Humanities Research Project	10 ECTS	Y		Mandatory	

---Technology, Ethics and Digital Society (10 ECTS)

[Data ethics: data collection_who owns data, who is excluded; data bias_gender bias, cultural bias]

[Algorithm governance: how algorithms impact society, recommendation systems, predictive systems; should algorithms be transparent?]

[AI and the humanities: AI writhing, AI art, AI historical analysis; has technology changed the concept of humanities knowledge?]

[Digital power structures: platform capitalism; digital labour; data colonialism]

---Digital Humanities Research Project (10 ECTS)

[Research pathways: students choose between two tracks

(1) Dissertation---independent research essay emphasizing critical analysis and theoretical engagement.
(2) Digital project---practice-based project applying digital tools to humanities research questions]

[Dissertation (Critical research): Students develop an original research question grounded in English, History, or Philosophy. Engagement with key debates in DH, such as data as constructed knowledge and algorithmic bias. Incorporate with critical frameworks.

Possible topics: limits of distant reading in literary studies; bias in digital archives and historical representation; the emergence of AI-generated texts challenges established concepts of knowledge and authorship.]

[Digital project (Applied DH practice): Students design and implement a DH project using appropriate tools. Projects must be research-driven (not purely technical) and grounded in a humanities question.

Possible formats: text analysis project (Python, NLP, topic modelling); digital archive or exhibition (Omeka, metadata design); spatial humanities project (GIS mapping, literary geography); network analysis (Gephi) --- working digital project + critical commentary]

[Critical reflection (required for both tracks) --- methodological choices and limitations; data selection and potential biases; relationship between digital tools and interpretation; ethical implications of the project]

[Students work under faculty supervision. Develop proposal, research design, and methodology. Participate in workshops on research methods, project designs, and academic writing if necessary. Emphasis on independent thinking and interdisciplinary integration.]

[Learning outcomes: Ability to formulate and investigate a humanities research question using digital methods. Critical understanding of the relationship between data, tools, and interpretation. Capacity to evaluate ethical and critical implications of digital research. Development of either advanced academic writing or applied digital research skills.]

This curriculum draft is designed around a progressive, interdisciplinary model of Digital Humanities education, informed by European undergraduate programs and leading international institutions. It responds to the current shift from early text encoding practices (e.g., XML/TEI) to data-driven and AI-oriented methodologies, ensuring students develop both technical competencies and critical awareness. Structurally, the program follows a progression from foundational concepts of data and digitization (Year 2), to methodological applications such as text mining and spatial humanities (Year 3), and finally to critical engagement with AI, ethics, and knowledge production (Year 4). The curriculum is explicitly aligned with core humanities disciplines, English, History, and Philosophy, by mapping specific digital methods to disciplinary research questions. A comparative analysis of DH programs in Ireland, the UK, Europe, and the US supports this design, as well as observations of technological clusters (e.g., NLP, GIS, metadata) across disciplines, ensuring both academic relevance and methodological coherence.

Gathering module information from official university websites across Ireland and Europe (e.g., UCC, Maynooth), collecting course titles, descriptions, ECTS weights, and disciplines into a structured dataset. This process involved manually identifying relevant Digital Humanities modules and extracting structured information to build a unified table of DH-related teaching provision. Then, using Python to process and analyze this dataset. The workflow followed by keyword extraction to identify recurring technical components (e.g.,

Python, NLP, GIS, metadata, machine learning). Conducting a co-occurrence analysis to examine how specific technologies cluster around different disciplines using the processed data. For example, text-based methods were strongly associated with linguistics and literary studies, while GIS and georeferencing appeared more frequently in history and cultural heritage contexts. Analyzing these results was critical in moving from raw data to curriculum design decisions. For instance, the identification of three dominant clusters, textual computation, spatial humanities, and data infrastructure, informed the structure of Year 3 modules.

The curriculum was then constructed by aligning these observed patterns with a progressive teaching model. These sequenced courses range from foundational concepts (data, digitization, computational thinking) to applied methods (text mining, GIS) and finally to critical and theoretical engagement (AI, ethics, philosophy of data). Also, matched technical skills to specific disciplinary contexts (English, History, Philosophy).